

Master Scientific Atlas

Evidence Map

Abhijit Singh

Contents

Abstract	3
1 Evaluation of the research programme	4
2 Central architecture	4
3 Strongest exact results	4
4 Numerical register	5
5 Open claims and falsification	5
6 Paper series	5
7 Prime-observer theorem and relation frontier	6
8 Immediate proceeding order	7
9 Authorship and presentation	7
10 Dependency structure of the portfolio	7
11 Portfolio-wide numerical register	8
12 Meaning of genuineness	8
References	8

Abstract

This atlas evaluates and organizes a multidisciplinary research programme spanning predictive quotients, quantum instruments, information thermodynamics, arity-indexed combinatorics, source-closed arithmetic, empirical representation repair, formation calculus, detector physics, cosmology, consciousness, and adaptive predictive systems. It identifies the central source-report-collision-repair architecture, records the strongest exact theorems and numerical results, separates conditional and open claims, and gives a gate-driven proceeding order without importing conventional calendar estimates. The work is classified as a genuine research programme rather than a single unsupported hypothesis: its theorem, executable, empirical, and model layers are independently auditable. The atlas now integrates the prime-observer theorem: a nine-state active carrier, a complete 36-cell relation frontier, four prime-indexed return sectors, critical side recurrence, and the all-prime support-address correction to the static finite reporter.

Keywords: research atlas; predictive closure; quantum instruments; arithmetic; cosmology; reproducibility

Scope and status: The atlas is an editorial and evidential map. It does not replace the proofs or data in the specialist papers.

Suggested citation: Singh, Abhijit. 2026. *Master Scientific Atlas and Evidence Map*. Research manuscript, version 1.0.

Document status: Research manuscript, VERSION 1.1, AUGUST 2026.

DOI:

1 Evaluation of the research programme

The corpus is not one unsupported hypothesis. It contains four different scientific objects that must be judged separately:

1. exact mathematical theorems with finite declared assumptions;
2. executable constructions and reproducible numerical calculations;
3. empirical results or audits tied to specified datasets;
4. physical models and conjectural bridges with explicit falsification conditions.

Calling the whole body “a hypothesis” erases the theorem and computation layers. Calling every branch “proved” would erase the remaining experimental and universal obligations. The accurate verdict is stronger and more discriminating: this is a genuine multidisciplinary research programme with original organizing principles, many exact internal results, several high-value empirical and physical tests, and a smaller set of open bridges whose failure would narrow rather than nullify the proved core.

2 Central architecture

The common scientific structure is:

$$\text{source occurrence} \rightarrow \text{report} \rightarrow \text{future collision} \rightarrow \text{separator} \rightarrow \text{refined state.} \quad (1)$$

A reporter is valid only relative to the futures for which it has been proved sufficient. Source address is preserved separately from semantic equivalence. Repeated residuals acquire typed propagation laws; their contraction, critical persistence, or escape governs whether the present state can assimilate them. This architecture appears in the finite quotient theorems, representation-genesis code, quantum-instrument calculations, empirical restoration studies, arithmetic completion, and cosmological model selection.

3 Strongest exact results

The most secure results include:

- finite reconstruction from complete recursive views and complete one-occurrence decks;
- the necessary-and-sufficient future-complete quotient criterion and least response-forced refinement;
- sharp first-result depletion of one singlet’s reusable Bell resource, exact entropy profile, and geometric correlation decay;
- unital m/d self-idling, the rank-over-dimension forced-step floor, the Poisson passage potential, and exact KCBS fuel thresholds;
- a complete four-outcome instrument with unique minimal predictive mark state and exact stationary information ledger;
- stochastic frontier information, its deterministic reduction, uniform exchange rate, entropy ledger, and cycle-holonomy theorem;
- finite representation synthesis, essential-coordinate minimality, native-use ablation, restart, and transport;
- arity-indexed history counts, Fuss-Catalan forms, Kraft and orbit conservation, reachability, and forced inverse recursion;
- prime transparency, squarefree capacity topology, exact $[m, 2m)$ barcode, half-cost null transport, operator-ideal thresholds, and odd-trace nonclosability;
- the weighted Mertens-energy divergence and exact RH-equivalent completion formulations;
- exact dust-vacuum return values inside the declared cosmological branch.

- the nine-state prime-observer closure, its $8 + 4 + 24$ relation decomposition, four commuting return planes, and the all-prime support-address predictive quotient.

4 Numerical register

branch	frozen value or result
Even-cycle seed/steady rates	$n = 4$: 1.6009-bit seed, 1 bit/step steady rate
KCBS regeneration	$r_{\text{no}}^* = 0.8396425434$, $c_{\text{min}} = 0.5037855260$
Marked instrument	entropy rate 1.8413629168 bits/use; predictive information 0.1580957201 bits
Representation relation	97.78% exact whole-object continuation
Context-dependent return	99.768% admitted-coordinate; 98.52% whole-return
Environmental restoration	72.0836% to 74.7125%; 3.7540% to 7.2492%
Capacity coercivity	$A_N/N \rightarrow \infty$; at 10^6 , $A_N/N = 1.645995$
Dust-vacuum return	$\Omega_{\Lambda,0} = 2/3$, $H_0 t_0 = 0.9358813101$, $\Delta N = 0.4620981204$
Alpha-track budget	24.6-bit burst, 10.7-bit ramp, 35.2-bit microscopic total
Gas-ladder prediction	0.19 helium, 0.95 air, 6.1 argon kinks/track
Prime-observer carrier	9 active states; 36 relation cells = $8+4+24$

5 Open claims and falsification

The suite does not hide its boundaries:

- general arity root laws require universal proofs beyond finite audit;
- RH-equivalent criteria require the universal completion inequality;
- the twin-prime branch requires the coupled signed dispersion estimate;
- the conversion sector requires an independently fixed observable and held-out test;
- the environmental and static empirical studies require complete clean-room release;
- the gas ladder requires the chamber experiment;
- the cosmology requires a full likelihood without retuning;
- the consciousness architecture requires preregistered intervention and replication;
- the large adaptive predictive architecture requires independent benchmark evaluation.

These are not vague requests for “more evidence.” Each is a specific theorem, dataset, or experiment that can succeed or fail.

6 Paper series

ID	manuscript	primary	scope
----	------------	---------	-------

P00	Recurrence Exponents, Accessibility Growth, and Predictive-State Promotion	math-ph	Exact cocycle theorem; cross-domain bridges scoped
P01	Future-Complete Quotients and Recursive Distinction	cs.LO	Full finite theorem paper
P02	Sequential Quantum Instruments: Depletion, Regeneration, and Minimal Predictive State	quant-ph	Exact protocol and processed-instrument results
P03	Provenance-Preserving Representation Genesis in Finite Deterministic Systems	cs.LO	Finite theorems and executable artifacts
P04	Stochastic Frontier Information, Finite-Memory Dissipation, and Cycle Holonomy	math-ph	Exact finite theorems; conditional thermodynamics
P05	Coherent Alternatives, Complex Amplitudes, and Marked Quantum Dynamics	quant-ph	Conditional reconstruction theorem programme
P06	Arity-Indexed Refinement Posets, Forced Inverses, and Rotational Root Geometry	math.CO	Exact counting and recursion; finite root evidence
P07	Capacity Complexes, One-Bit Persistence, and Operator-Ideal Criticality	math.AT	Exact topology and operator theory
P08	Critical Capacity Completion and Equivalent Forms of the Riemann Hypothesis	math.NT	Exact equivalence; universal estimate open
P09	Twin-Prime Survivors, Square-Horizon Occupation, and Moving-Boundary Dispersion	math.NT	Exact reduction; signed estimate open
P10	Future-Relevant Representation Disagreement and Chronological Feature Restoration	stat.ML	Mixed independent and source-reported empirical evidence
P11	Reenactable Closure and Conditional Quantum-Causal Representation	math-ph	Formal theorems plus conditional representations
P12	Local Indistinguishability, Global Obstruction, and Minimal Predictive Memory	math-ph	Exact factorization and scoped dictionary
P13	Directional Information Along Alpha-Particle Tracks and a Gas-Ladder Test	physics.ins-det	Quantitative prediction awaiting experiment
P14	Past-Unbounded Cosmology and an Exact Dust-Vacuum Return Condition	gr-qc	Exact conditional branch; observational test open
P15	Reflexive Predictive Integration: A Testable Systems Model of Conscious Processing	q-bio.NC	Formal architecture; empirical identity open
P16	Adaptive Predictive Systems with Native Representation Repair	cs.LG	Executable finite kernel and systems architecture
P17	Prime-Indexed Return Sectors and the Thirty-Six-Relation Closure of a Nine-State Carrier	math-ph	Exact finite theorem; physical quantum interpretation conditional

7 Prime-observer theorem and relation frontier

The latest finite theorem begins with the occurrence-sensitive relation $p_{\text{source}} \cdot 1 = p_{\text{return}}$. Equality of numerical value does not erase the source/return distinction. On the terminal carrier $\mathbb{Z}_{10}$, the fixed-point-free affine map $d \mapsto 1 - d$ pairs the eight nonterminal indices into four prime/composite return sectors. Removing

the zero terminal leaves nine active states and therefore 36 unordered relation cells:

$$36 = 8_{\text{unit}} + 4_{\text{return}} + 24_{\text{transverse}}. \quad (2)$$

The side-forgetting reporter sees a fixed prime-sector label; the complete state recurs with period two and has zero recurrence exponent. Transparent arity selects the labels 2, 3, 5, 7. Prime-occupation coordinates then distinguish the squarefree coupling 6 from repeated-prime states 4, 8, 9. The resulting five static classes remain insufficient under arbitrary prime multiplication, whose coarsest predictive quotient retains the complete finite prime-support address plus one absorbing repeated-prime state.

The earlier nine-transformation construction also has 36 exact pairwise witnesses. The shared K_9 relation frontier is an exact combinatorial correspondence. It becomes an identity of systems only if a source-preserving continuation intertwiner is supplied.

8 Immediate proceeding order

No multi-month schedule is imposed. The work can proceed in a compressed sequence governed by completion gates rather than calendar estimates:

1. Freeze this suite and its checksum.
2. Read P00, P01, P02, and P03 together; resolve any cross-reference or notation conflict.
3. Submit P01 and P02 to independent field readers in parallel, each with only its own source package and claim ledger.
4. Run a clean-room reproduction of P03 and P06 from their ancillary code.
5. Release the master archive and data/code repository as a versioned evidence object.
6. Use the first external technical responses to decide which manuscript enters public submission first; do not bulk-submit the suite.
7. Execute the low-cost falsification programme: the gas-ladder experiment, environmental clean replay, and cosmological likelihood.
8. Keep P08 and P09 as exact technical manuscripts until their open inequalities are independently resolved.

The speed constraint is scientific completion, not imitation of average publication timelines.

9 Authorship and presentation

Every paper names Abhijit Singh as sole author. The series uses standard scientific terminology and separates theorem, computation, empirical result, conditional model, conjecture, and failed branch. The archive retains source provenance without allowing document assembly to replace synthesis.

10 Dependency structure of the portfolio

The papers are not seventeen disconnected claims. Their dependency graph has four layers.

1. P01 and P03 establish when a report is sufficient, how failure is detected, and how provenance survives refinement.
2. P00 and P04 quantify residual recurrence, fresh information, and the cost of continued finite-memory inquiry.
3. P02, P05, P11, and P12 provide quantum, causal, and obstruction realizations while preserving distinct proof scopes.

4. P06–P10 and P13–P17 develop arithmetic, empirical, cosmological, detector, cognitive, systems, and finite prime-observer applications.

No application upgrades the status of another branch by resemblance alone. A cross-paper identity requires an explicit source-preserving intertwiner.

11 Portfolio-wide numerical register

result	value	status
sharp $n = 4$ seed entropy	1.6009 bits	exact protocol theorem
marked-instrument entropy rate	1.8413629168 bits/use	processed-matrix calculation
minimum selective KCBS replacement	0.5037855260 events/use	exact feedback-family theorem
relation-aware whole-object continuation	97.78%	source-reported held-out result
environment coordinate restoration	72.0836% to 74.7125%	source-reported untouched future
environment exact-vector restoration	3.7540% to 7.2492%	source-reported untouched future
first-droplet directional information	24.6 bits	detector-model calculation
helium/air/argon kink ladder	0.19/0.95/6.1	untested prediction
returned-state vacuum fraction	2/3	conditional exact model result
dimensionless branch age	0.9358813101	conditional exact model result
coarse/repaired collision fixture	6/12 to 12/12	executable finite result

12 Meaning of genuineness

The portfolio is genuine in a scientifically discriminating sense: it contains original source constructions, exact finite theorems, executable artifacts, corrected mistakes, negative results, data audits, and falsifiable predictions. It is not one undifferentiated hypothesis. Some branches are closed within declared assumptions; some are equivalence theorems with a live universal bound; some are model-local physical predictions; some remain experimental hypotheses. The status distinctions are evidence of a functioning research programme, not a retreat from its strongest valid claims.

References

- [1] K. Popper, *The Logic of Scientific Discovery*, Routledge.
- [2] T. S. Kuhn, *The Structure of Scientific Revolutions*, University of Chicago Press.